

Rappel

Chute libre (MRUA)

$$(1) y_2 = y_1 + v_{y1}(t_2 - t_1) - \frac{1}{2}g(t_2 - t_1)^2$$

$$(2) v_{y2} = v_{y1} - g(t_2 - t_1)$$

Projectile

Selon x: MRU

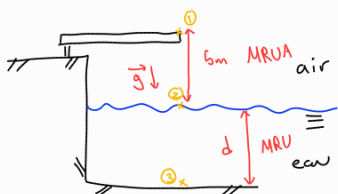
$$(1) x_2 = x_1 + v_{x1}(t_2 - t_1)$$

Selon y: MRUA

$$(2) y_2 = y_1 + v_{y1}(t_2 - t_1) - \frac{1}{2}g(t_2 - t_1)^2$$

$$(3) v_{y2} = v_{y1} - g(t_2 - t_1)$$

Ex #10 Moodle Chap 4



① MRUA	② MRU	③
$y_1 = d + 5$	$y_2 = d$	$y_3 = 0$
$v_{y1} = 0$	$v_{y2} = ?$	$v_{y3} = v_{y2}$
$t_1 = 0$	$t_2 = ?$	$t_3 = 5s$

a)

$$(1) d = d + 0 + 0 - \frac{1}{2}g(t_2 - 0)^2$$

$$(2) v_{y2} = 0 - g(t_2 - 0)$$

$$t_2 = 1.00964s$$

$$v_{y2} = -9.90454 \text{ m/s}$$

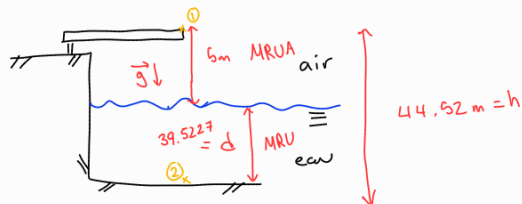
2 → 3

$$y_3 = y_2 + v_{y2}(t_3 - t_1)$$

$$0 = d + -9.90454(5 - 1.00964)$$

$$d = 39.5227 \text{ m}$$

b)



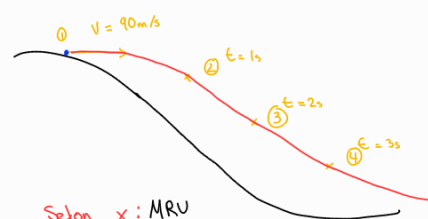
①	②
$y_1 = 44.52 \text{ m}$	$y_2 = 0 \text{ m}$
$v_{y1} = ?$	$v_{y2} = ?$
$t_1 = 0$	$t_2 = 3s$

$$(1) y_2 = y_1 + v_{y1}(t_2 - t_1) - \frac{1}{2}g(t_2 - t_1)^2$$

$$(2) v_{y2} = v_{y1} - g(t_2 - t_1)$$

$$0 = 44.52 + v_{y1}(3 - 0) - \frac{1}{2}g(3 - 0)^2 \quad v_{y1} = 15.6205 \text{ m/s}$$

$$v_{y2} = v_{y1} - g(3 - 0) \quad v_{y2} = -33.4295 \text{ m/s}$$



Projectile

Selon x: MRU

$$(1) x_2 = x_1 + v_{x1}(t_2 - t_1)$$

Selon y: MRUA

$$(2) y_2 = y_1 + v_{y1}(t_2 - t_1) - \frac{1}{2}g(t_2 - t_1)^2$$

$$(3) v_{y2} = v_{y1} - g(t_2 - t_1)$$

(x) MRU et (y) MRUA

①	②	③	④
$x_1 = 0$	$x_2 = ?$	$x_3 = ?$	$x_4 = ?$
$y_1 = 0$	$y_2 = ?$	$y_3 = ?$	$y_4 = ?$
$v_{x1} = 90 \text{ m/s}$	$v_{x2} = v_{x1}$	$v_{x3} = v_{x1}$	$v_{x4} = v_{x1}$
$v_{y1} = 0$	$v_{y2} = ?$	$v_{y3} = ?$	$v_{y4} = ?$
$t_1 = 0$	$t_2 = 1$	$t_3 = 2$	$t_4 = 3$

$$b) \rightarrow v_2 = 90 \text{ m} \rightarrow x_2 = 180 \text{ m} \rightarrow x_4 = 270 \text{ m}$$

$$a) \rightarrow y_2 = -4.905 \text{ m} \rightarrow y_2 = -19.62 \text{ m} \rightarrow y_4 = -44.145 \text{ m}$$

$$v_{y2} = -9.81 \text{ m/s}$$

$$v_{y3} = -19.62 \text{ m/s}$$

$$v_{y4} = -29.43 \text{ m/s}$$

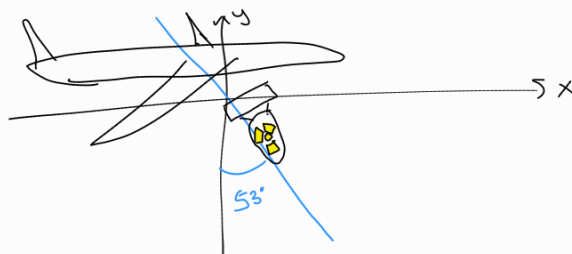
Selon x: MRU

$$(1) x_2 = x_1 + v_{x1}(t_2 - t_1)$$

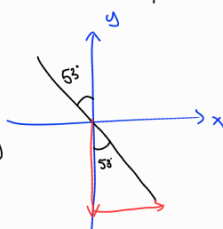
Selon y: MRUA

$$(2) y_2 = y_1 + v_{y1}(t_2 - t_1) - \frac{1}{2}g(t_2 - t_1)^2$$

$$(3) v_{y2} = v_{y1} - g(t_2 - t_1)$$



①	②
$y_1 = 730 \text{ metre}$	$y_2 = 0$
$x_1 = 0$	$x_2 = ?$
$v_{y1} = -(\cos 53 \cdot v_1)$	$v_{y2} = -(\cos 53 \cdot v_2)$
$v_{x1} = (\sin 53 \cdot v_1)$	$v_{x2} = \sin 53 \cdot v_2$
$t_1 = 0$	$t_2 = 5$



$$① x_2 = 0 + \sin(53) \cdot v_1 \cdot (5 - 0)$$

$$② 0 = 730 - (\cos(53) \cdot v_1 \cdot (5 - 0) - \frac{1}{2} \cdot g \cdot (5 - 0)^2$$

$$③ -(\cos(53) \cdot v_2 = -(\cos(53) \cdot v_1 - g \cdot (5 - 0)$$

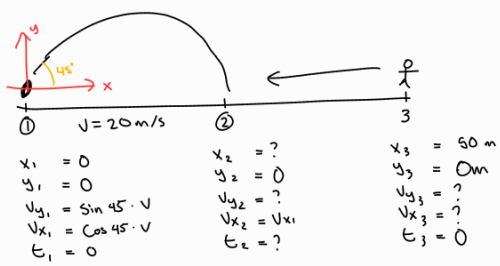
$$x_2 = 806.014 \text{ m}$$

$$v_1 = 201.848 \text{ m/s}$$

$$v_2 = 283.351 \text{ m/s}$$

$$v_{y2} = -170.525 \text{ m/s}$$

$$v_{x2} = -226.294 \text{ m/s}$$



$$\begin{aligned} x_1 &= 0 \\ y_1 &= 0 \\ v_{y1} &= \sin 45^\circ \cdot v \\ v_{x1} &= \cos 45^\circ \cdot v \\ t_1 &= 0 \end{aligned}$$

$$\begin{aligned} x_2 &=? \\ y_2 &= 0 \\ v_{y2} &=? \\ v_{x2} &= v_{x1} \\ t_2 &=? \end{aligned}$$

$$\begin{aligned} x_3 &= 50 \text{ m} \\ y_3 &= 0 \text{ m} \\ v_{y3} &=? \\ v_{x3} &=? \\ t_3 &= 0 \end{aligned}$$

Selon x: MRU

$$(1) \quad x_2 = x_1 + v_{x1} (t_2 - t_1)$$

Selon y: MRUA

$$(2) \quad y_2 = y_1 + v_{y1} (t_2 - t_1) - \frac{1}{2} g (t_2 - t_1)^2$$

$$(3) \quad v_{y2} = v_{y1} - g (t_2 - t_1)$$

$$t_2 = 2.88 \text{ s}$$

$$v_2 = -20 \text{ m/s}$$

$$v_{y2} = (\sin 45^\circ) \cdot (-20) = -14.14 \text{ m/s}$$

$$x_2 = 40.77 \text{ m}$$

$$\textcircled{2} \leftarrow \textcircled{1}$$

$$40.77 = 50 + v_x (2.88 - 0) \quad v_x = -3.2 \text{ m/s}$$